

RMO(INDIA) – 2007

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let $\triangle BAC$ be an acute angled triangle; AD is the bisector of $\angle BAC$ with D on BC , and BE be the altitude from B to AC . Show that $m\angle CED > 45^\circ$. [17]
 2. Let a, b, c be three natural numbers such that $a < b < c$ and $\gcd(c-a, c-b) = 1$. Suppose there exist an integer d , such that $a+d, b+d, c+d$ form the sides of a right angled triangle. Prove that there exist integers l, m such that $c+d = l^2 + m^2$. [17]
 3. Find all pairs (a, b) of real numbers such that whenever α is a root of $x^2 + ax + b = 0$, $\alpha^2 - 2$ is also a root of that equation. [17]
 4. How many 6 digits numbers are there such that: [16]
 - (i) The digits of each number are all from the set $\{1, 2, 3, 4, 5\}$.
 - (ii) Any digits that appears in the number appears at least twice?
 5. A trapezium $ABCD$, in which AB is parallel to CD , is inscribed in a circle with centre O . Suppose the diagonals AC and BD of the trapezium intersect at M and $OM = 2$. [17]
 - (i) If $m\angle AMB = 60^\circ$, find with proof, the difference between the lengths of the parallel sides.
 - (ii) If $m\angle AMD = 60^\circ$, find with proof, the difference between the lengths of the parallel sides.
 6. Prove that: [16]
 - (i) $5 < \sqrt{5} + \sqrt[3]{5} + \sqrt[4]{5}$.
 - (ii) $8 > \sqrt{8} + \sqrt[3]{8} + \sqrt[4]{8}$
 - (iii) $n > \sqrt{n} + \sqrt[3]{n} + \sqrt[4]{n}$ for all integers $n \geq 9$.

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